The DFT Approximation to the FT

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Key Concepts

1) The DFT is frequently used for numerical spectral analysis of continuous-time signals.

2) Three factors affect the accuracy of the DFT approximation to the Fourier transform (FT).

    a) The signal must be sampled in time.

    b) The duration of the signal must be finite or truncated.

    c) The DFT can only calculate spectra at discrete values of frequency.

3) Sampling in time corresponds to approximating the integral in the FT with a Riemann sum.

    a) The sampling theorem provides guidance on choosing the sampling interval.

    b) The quality of the approximation improves as the sampling interval decreases.

4) Truncation of the signal to finite duration corresponds to multiplying the signal by a window.

    a) The window is one for the desired signal span and zero otherwise.

b) Truncation corresponds to convolving the sampled signal discrete-time Fourier transform (DTFT) with the DTFT of the window.

c) The DTFT of the window is a ratio of sines and has a main lobe centered at zero surrounded by sidelobes.

d) Convolution with the window DTFT blurs the desired DTFT.

5) The DFT samples the DTFT associated with the windowed, sampled signal at uniformly spaced frequency intervals.

a) If the DFT is length M, the frequency samples are $\omega_k = k\frac{2\pi}{M}$ rads or $\Omega_k = k\frac{2\pi}{MT}$ rads/sec.

b) Increasing the DFT length by padding the signal with zeros does not overcome the blurring introduced by the truncation step. The DFT is merely sampling the underlying DTFT which is already blurred.

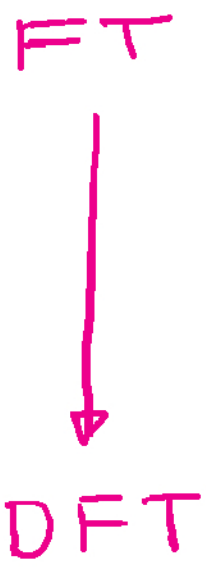
Video Screenshots

Using the DFT to Approximate the FT

- Numerical analysis of the frequency content of continuous-time signals

Three Approximation Steps

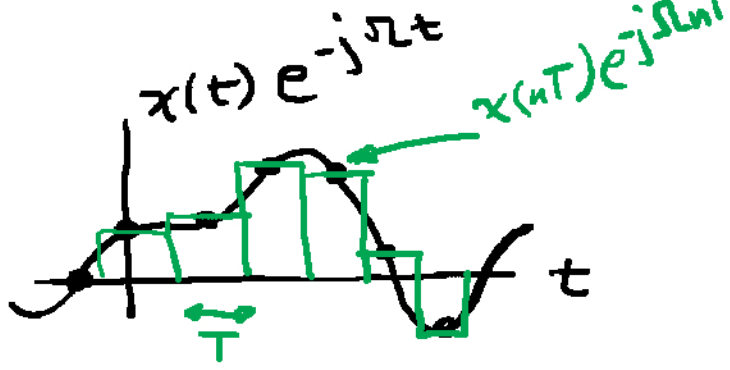
1) Sample in time
2) Truncate in time
3) Sample in frequency



FT: $$X(\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t}\, dt$$

1) Sample in time: Riemann sum approximation

$$\hat{X}(\Omega) = \sum_{n=-\infty}^{\infty} x(nT)\, e^{-j\Omega nT}\, T$$



$$= T \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\Omega Tn}$$

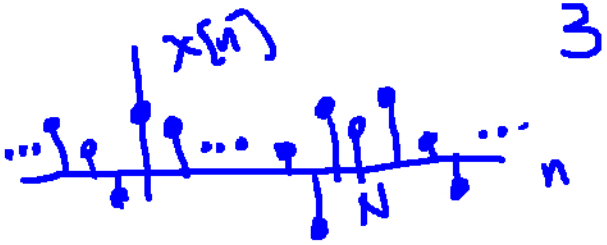
$$= T\, X(e^{j\omega})\Big|_{\omega=\Omega T}$$ DTFT

Accuracy: a) Sampling Theorem

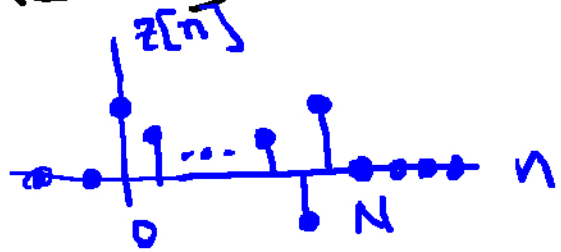
b) Approx improves as $T \downarrow$

2) Truncate to N samples

$$\hat{X}_N(\Omega) = T \sum_{n=0}^{N-1} x[n] e^{-j\Omega T n}$$



Let $z[n] = \begin{cases} x[n], & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}$ $\xleftrightarrow{\text{DTFT}}$ $Z(e^{j\omega})$



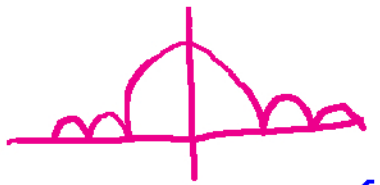
$$\hat{X}_N(\Omega) = T\, Z(e^{j\omega})\big|_{\omega = \Omega T}$$

But $z[n] = x[n]w[n]$ where $w[n] = \begin{cases} 1, & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases}$

$\updownarrow$ DTFT

$$W(e^{j\omega}) = e^{-j(N-1)\omega/2} \frac{\sin\left(\frac{N\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}$$

$$Z(e^{j\omega}) = \frac{1}{2\pi} X(e^{j\omega}) * W(e^{j\omega})$$



distorted version of $X(e^{j\omega})$

3) Sample frequency at $\omega_k = \frac{2\pi}{M} k$ $(M \geq N)$

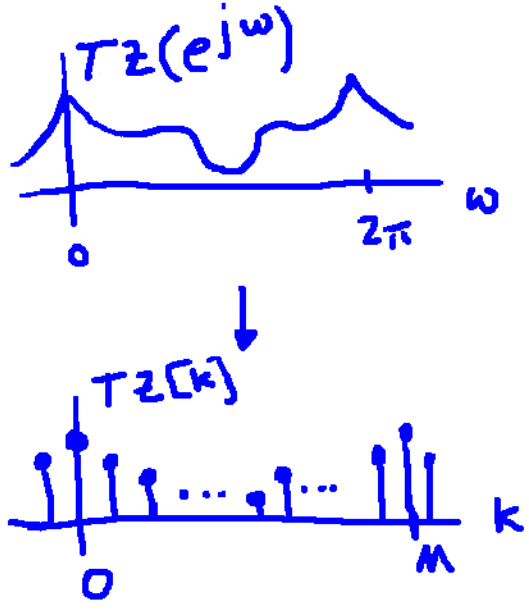
$$\Omega_k = \frac{2\pi}{MT} k$$

$$\hat{X}_N(\Omega_k) = T \sum_{n=0}^{N-1} x[n] e^{-j\left(\frac{2\pi}{MT}\right) kTn}$$

$$= T \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{M} kn}$$

$$= T \; Z(e^{j\omega}) \Big|_{\omega_k = \frac{2\pi}{M} k}$$

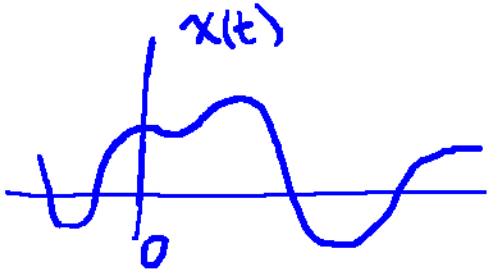
$$= T \; Z[k] \quad \text{where} \quad z[n] \xleftrightarrow{DFT;M} Z[k]$$



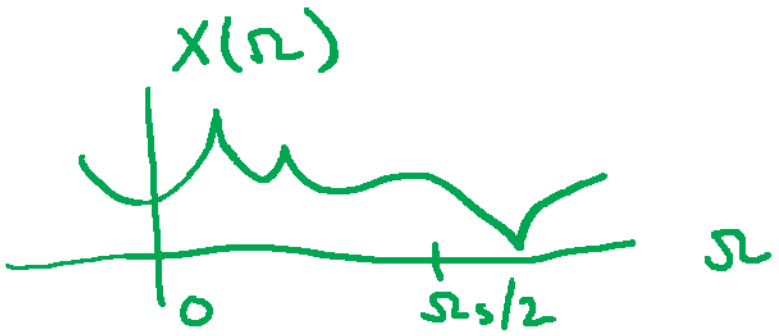
Increase M (zero padding) $\Rightarrow$ sample $Z(e^{j\omega})$ more densely

Does not eliminate distortion due to $W(e^{j\omega})$

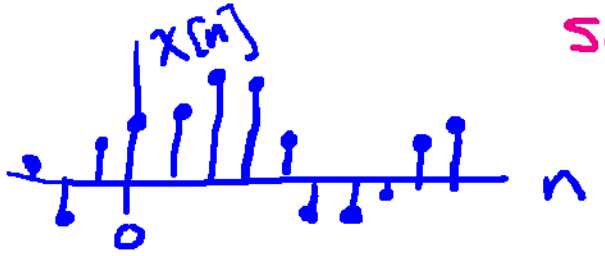
Summary



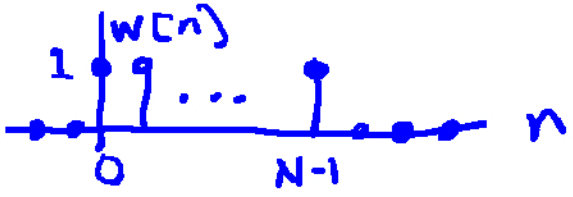
FT



sampling ⇒ limits bandwidth

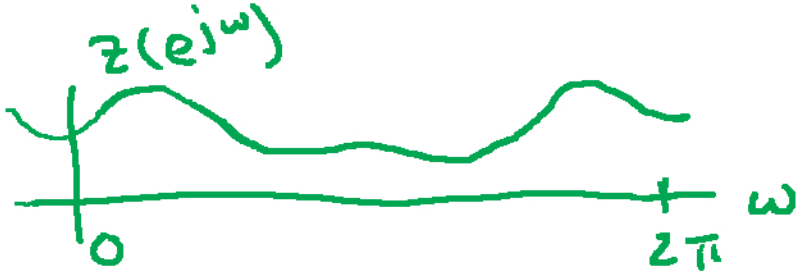


x

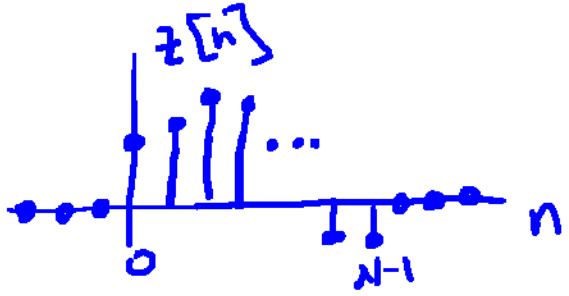


truncation/windowing

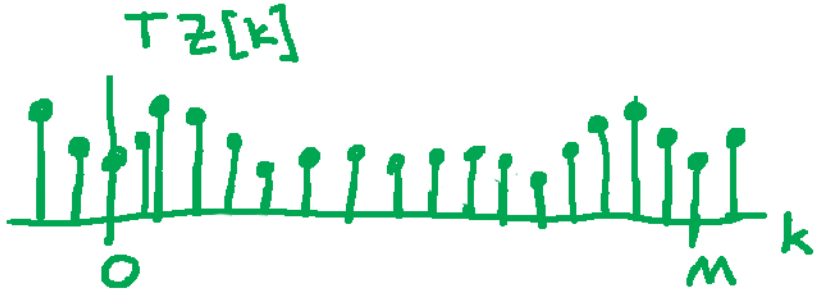
⇒ smoothing/distortion



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DFT; M



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